CDS535 Data Mining

Comparative Data Mining with Mushroom and Ionosphere Datasets

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Group 9

Overview

/ C O N T E N T S

1. Ionosphere Data Analysis and Preprocessing
   ZENG Xinying
2. Mushroom Data Analysis and Preprocessing
   WU Yanping
3. Classification Models and Association Rules
   HU Ziqi
4. Results and Cross-Dataset Evaluation
   ZHANG Jiaming
5. Discussion, Limitations and Future Work
   HU Sirui

PART ONE

Dataset Analysis

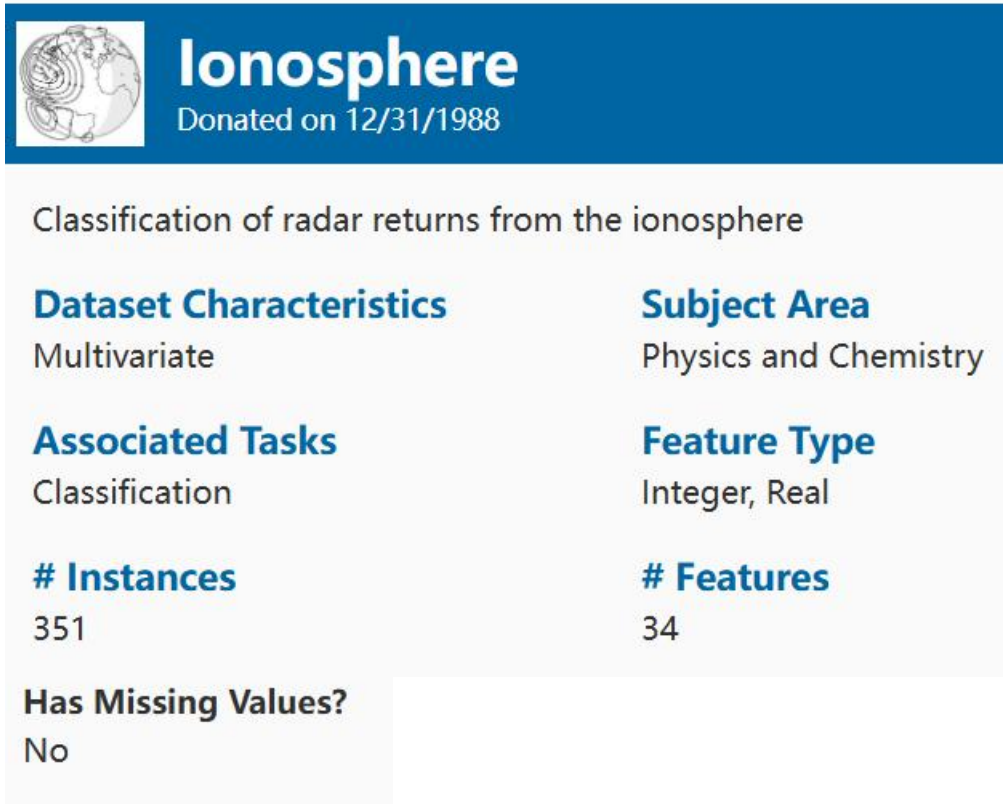
ionosphere



Dataset A: Ionosphere

Dataset Overview

Total Instances: 351 radar records
Total Attributes: 35 (34 numerical predictors and 1 target class)
Data Type: Numerical predictors and one nominal target attribute
Classification Task: Binary classification
Missing Values: 0



Target Variable: class

Bad (b): 126 instances (35.9%)
Good (g): 225 instances (64.1%)

The dataset has a moderately imbalanced class distribution, with class g (good returns) being more common than class b (bad returns).

Data Visualisation & Class Distribution

Selected attribute
Name: a03 Type: Numeric
Missing: 0 (0%) Distinct: 219 Unique: 216 (62%)

Statistic	Value
Minimum	-1
Maximum	1
Mean	0.641
StdDev	0.498

Class: class (Nom) Visualize All

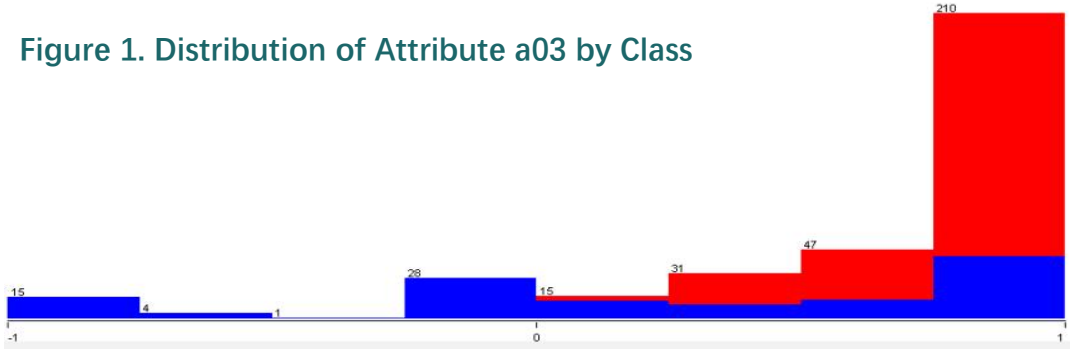


Figure 1. Distribution of Attribute a03 by Class

Key Observations

1. Class Imbalance
Class g (64.1%) is more common than class b (35.9%).
2. Feature Distribution
Attribute a03 shows different distributions across the two classes.
3. Class Overlap
The scatter plot shows partial overlap between the two classes.

Selected attribute
Name: class Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	b	126	126
2	g	225	225

Class: class (Nom) Visualize All



Figure 2. Class Distribution of the Ionosphere Dataset

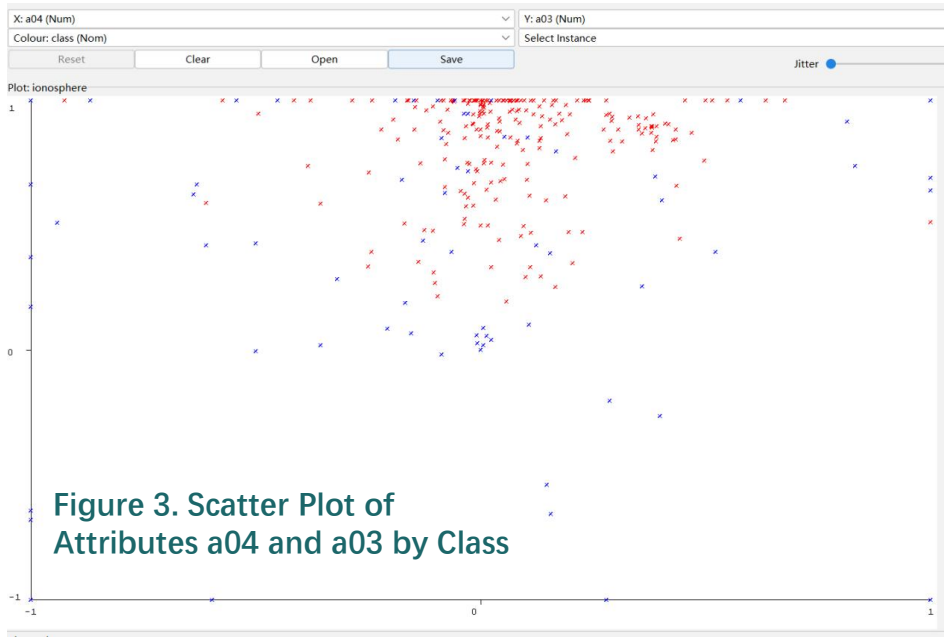


Figure 3. Scatter Plot of Attributes a04 and a03 by Class

Data Preprocessing Comparison

Original Dataset vs. Normalized Dataset

Experimental Settings

Classifier: RandomForest

Filter: Normalize

Evaluation: 10-fold Cross-validation

FilteredClassifier: Prevent data leakage

Experimental Results

```
Correctly Classified Instances         326               92.8775 %
Incorrectly Classified Instances        25                7.1225 %
```

```
Correctly Classified Instances         325               92.5926 %
Incorrectly Classified Instances        26                7.4074 %
```

Metric	Original	Normalize
Accuracy	92.88%	92.59%
Weighted F1	0.928	0.925

Normalization did not improve model performance.

Model Results & Interpretation

/ RandomForest Results

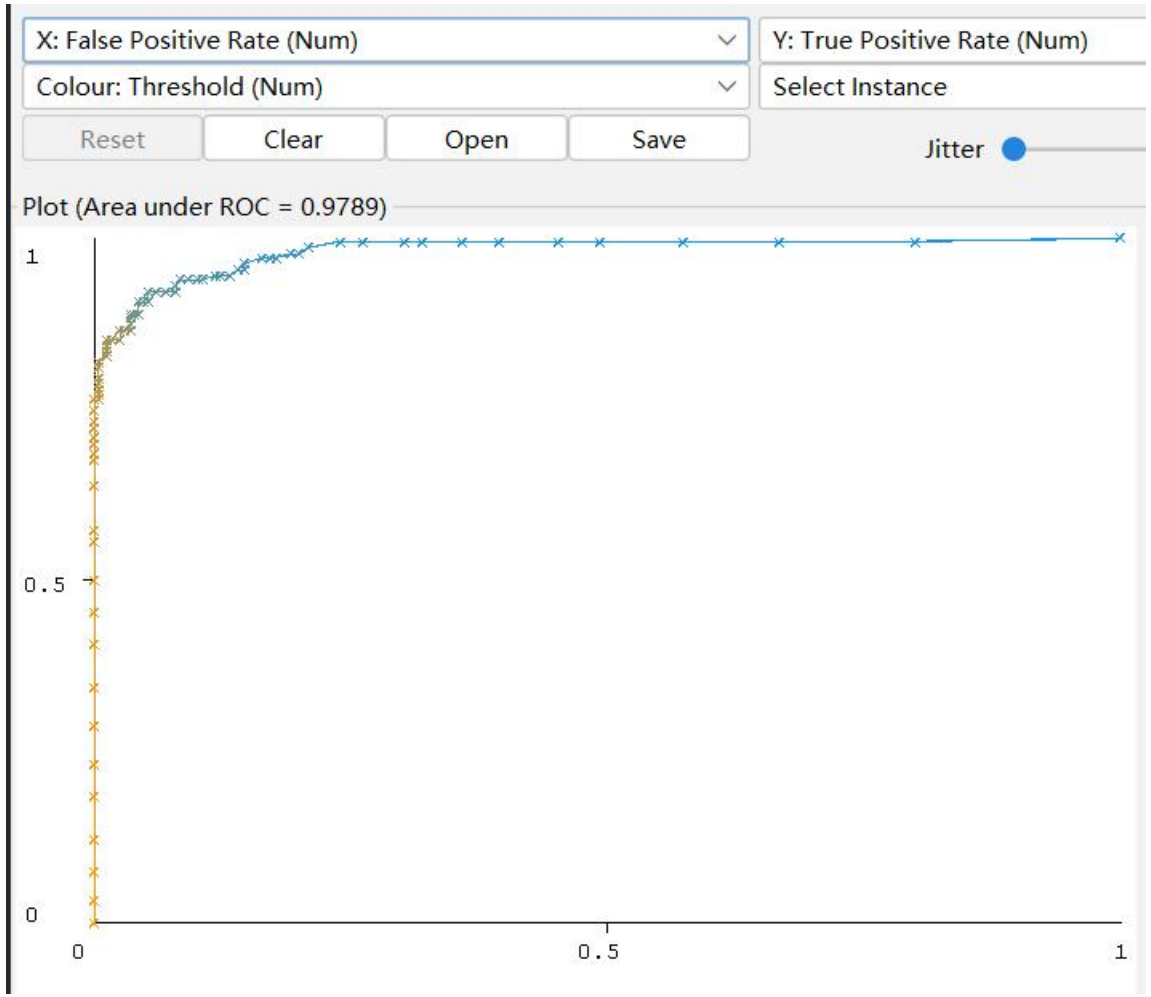


Figure 4. RandomForest ROC Curve (AUC = 0.9789).

Correctly Classified Instances	326	92.8775 %
Incorrectly Classified Instances	25	7.1225 %
Kappa statistic	0.8428	
Mean absolute error	0.1287	
Root mean squared error	0.2255	
Relative absolute error	27.951 %	
Root relative squared error	47.0057 %	
Total Number of Instances	351	

10-fold cross-validation

Correct predictions: 326 / 351

Accuracy: 92.88% — highest among four models

ROC-AUC: 0.9789 for class b

Compared with J48 (Experimenter)

Accuracy gain: not significant

AUC gain: significantly higher than J48's

PART TWO

Dataset Overview

Mushroom



Dataset B: Mushroom

Dataset Overview

Total Instances: 8,124 mushroom records

Total Attributes: 23 (22 descriptive attributes and 1 target class)

Data Type: All attributes are nominal (categorical)

Classification Task: Binary classification



Target Variable: class

Edible (e): 4,208 instances (51.77%)

Poisonous (p): 3,916 instances (48.23%)

The dataset has a relatively balanced class distribution, with neither class significantly dominating the other.

Dataset B: Mushroom

NO-01

Key Features

The dataset describes mushrooms using characteristics such as:

1.Cap shape, surface, and color

2.Odor

3.Gill attachment, spacing, size, and color

4.Stalk characteristics

5.Ring type, spore-print color, population, and habitat

Project Objective

To investigate the Mushroom dataset and classify mushrooms as edible or poisonous based on their categorical characteristics, while examining data quality issues and the effects of preprocessing strategies on classification performance.

NO-02

Data Quality Analysis



Data Quality Analysis

Missing Values

Attribute: stalk-root

Missing values: 2,480 (30.53%)

Other attributes: No missing values



Constant Attribute

Attribute: veil-type

Only one distinct value appears in the dataset.It provides no useful information for distinguishing the classes.

Data Cleaning



Data Quality Analysis

Remove Constant Attribute

Weka Filter: Remove

Attribute removed: veil-type

Reason: Only one distinct value.



Handle Missing Values

Weka Filter: RemoveWithValues

Attribute: stalk-root

Setting: matchMissingValues = True

Action: Remove instances with missing values.

PART THREE

Classification Modeling

Mushroom and Ionosphere

Four classifiers, training-only preprocessing and Apriori



Algorithms and Parameters

MODELING METHODS

Model comparison with fixed settings

Experimental Design

Mushroom: 5,644 complete records

Ionosphere: 351 original records

10-fold CV and 66%/34% split

Fixed parameters and split seed = 1



J48

Pruning confidence factor: 0.25

Minimum instances per leaf: 2

RandomForest

100 trees, model seed = 1

Tree-count runs: 50 / 100 / 200

Logistic and SMO

Logistic: ridge = 1.0E−8

SMO: C = 1, linear kernel

Workflow and Leakage Prevention

PIPELINE AND WEKA DEMO

Mushroom

Drop missing rows and veil-type

21 nominal predictors

No external standardization

Ionosphere

J48 / RF: original predictors

Logistic / SMO: scale within training folds

SMO internal scaling: Neither

Feature Selection

CfsSubsetEval + BestFirst

Nested with J48, select on training data

Apply the same subset to test data

WEKA Workflow Demo

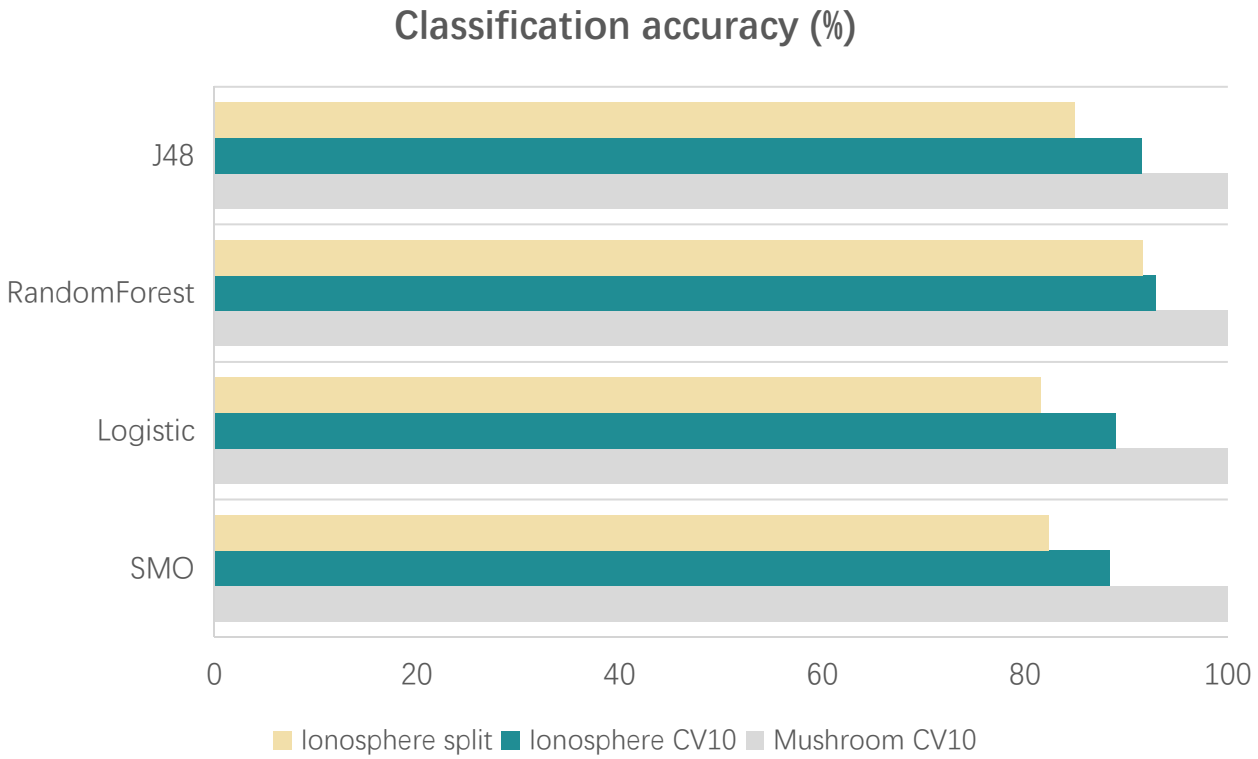
Standardize inside FilteredClassifier

Choose Logistic or SMO, run 10-fold CV

Inspect confusion matrix and positive ROC

Results and Association Rules

CLASSIFICATION AND ASSOCIATION



Classification Results

Mushroom: all four models at 100%

Ionosphere: RF has highest accuracy

CV: 92.88%, split: 91.60%

Accuracy differences are not significant

Apriori Association Rules

Remove class, obtain 20 rules

Example support: 97.236%

Confidence: 100%, lift: 1.003

Frequent consequent drives high confidence

PART FOUR

Results & Evaluation

I led model evaluation and the cross-dataset comparison: two protocols, four classifiers, statistical significance testing, and Apriori review.



My Role & Evaluation

/ ROLE & PROTOCOLS

As the evaluation lead, I applied the same evaluation design to every classifier so the results are comparable and reproducible.

2 Protocols

10-fold stratified cross-validation and an independent 66%/34% split, seed fixed to 1.

6 Metrics

Accuracy, precision, recall, F1-score, ROC-AUC and Kappa; rules use support, confidence and lift.

Leakage Check

Data-driven preprocessing is fitted inside each fold, consistent with the modeling pipeline.

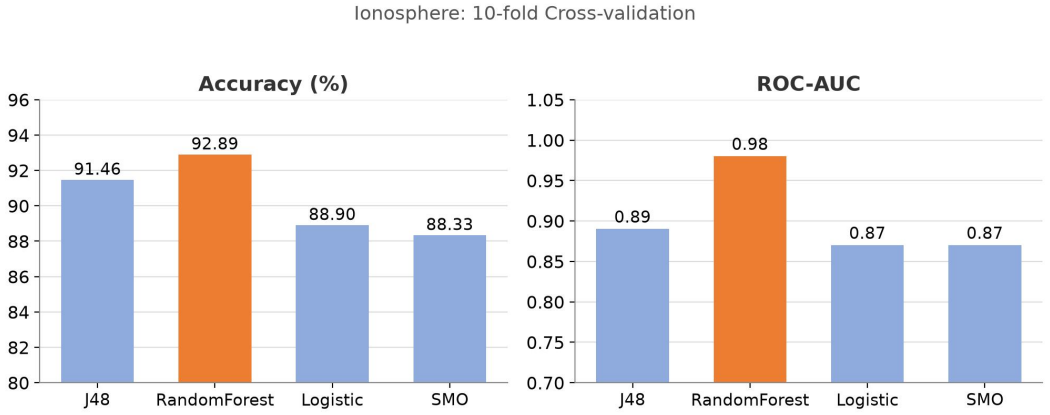
4 Classifiers

J48, RandomForest, Logistic and SMO evaluated on Mushroom and Ionosphere.

Quantitative Results

/ 10-FOLD CROSS-VALIDATION

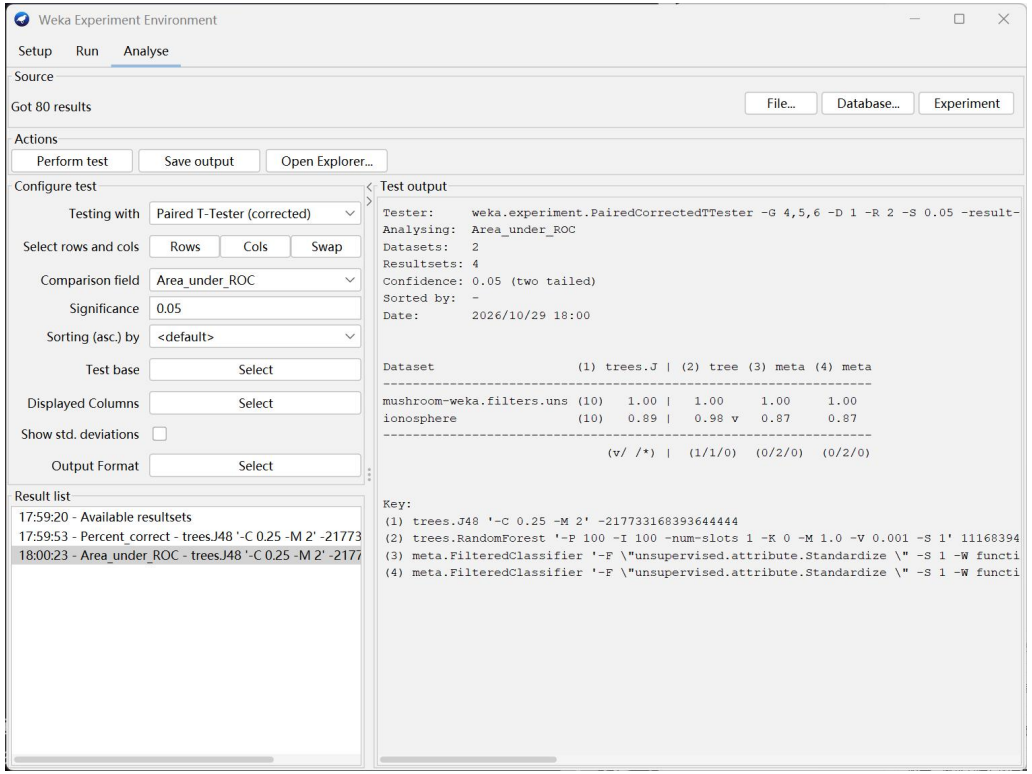
Model	Mushroom Acc (%)	Ionosphere Acc (%)	Ionosphere AUC
J48	100.00	91.45	0.89
RandomForest	100.00	92.88	0.98
Logistic	100.00	88.89	0.87
SMO	100.00	88.32	0.87
Best	All tie	RandomForest	RandomForest



RandomForest leads on Ionosphere (92.88%, AUC 0.98), while every classifier reaches 100% on Mushroom.

Model Comparison & Significance

/ PAIRED CORRECTED T-TEST, 0.05



Weka Experimenter output: ROC-AUC across the ten folds

What the test shows

- **Accuracy: no significant gap**

  All classifiers return 0/2/0 against J48, so the accuracy differences are not statistically significant.

- **AUC: RandomForest is significant**

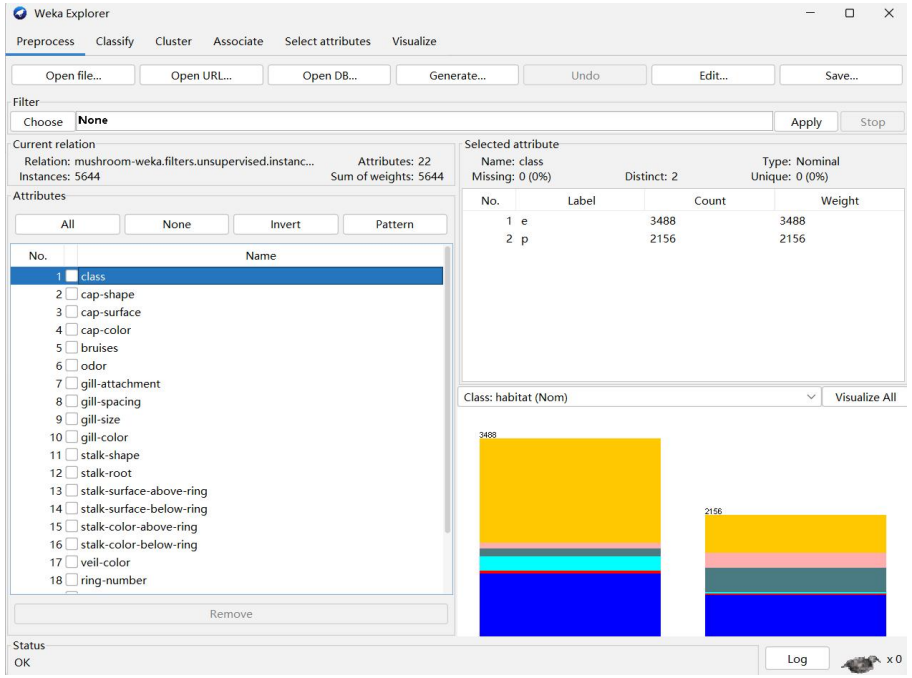
  RF records 1/1/0 and is significantly better than J48 on ROC-AUC.

- **Careful wording**

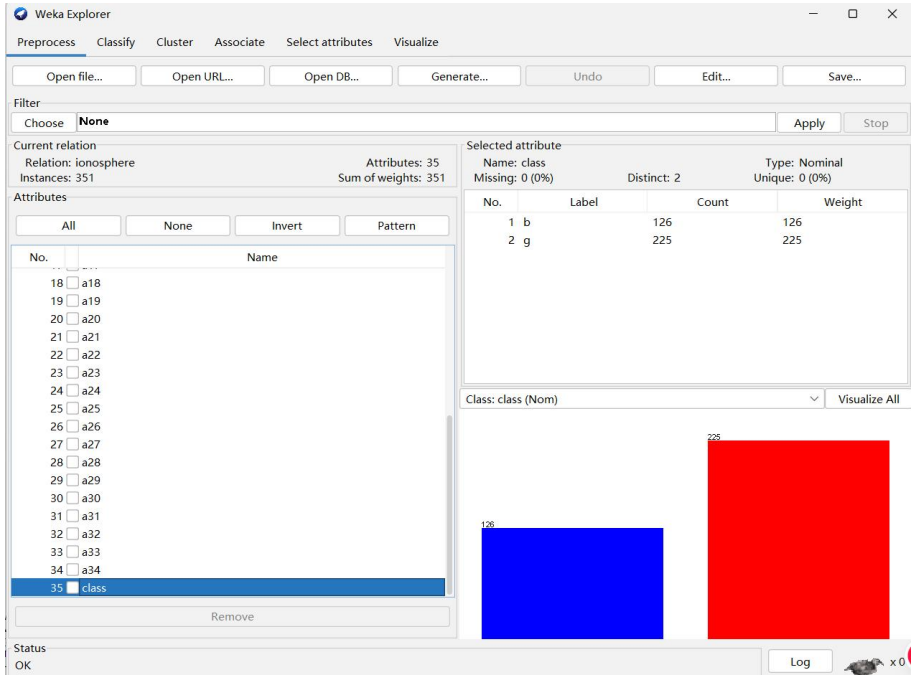
  "Highest observed" is not the same as "significantly best"; significance is confirmed only for RF's AUC.

Cross-Dataset Comparison

/ MUSHROOM vs IONOSPHERE



Mushroom: 5,644 instances, 22 attributes



Ionosphere: 351 instances, 35 attributes

Mushroom hits a metric ceiling (all 100%), while Ionosphere differentiates the models. They differ in scale, attribute type, sample retention and parameters, so cross-dataset gaps cannot be attributed to any single factor.

Key Findings

/ SUMMARY & TAKEAWAYS

RandomForest leads on Ionosphere

Highest accuracy (92.88%) and a significantly higher ROC-AUC (0.98).

Mushroom is a perfect tie

All four classifiers reach 100% accuracy and AUC 1.00, so we report a tie rather than ranking them.

Added complexity is not always useful

Neither feature selection nor more trees gave a stable improvement.

Ensembles pay off on noisy data

The ensemble benefit appears only on the noisier, continuous dataset, not on the already-separable nominal one.

PART Five

Discussion & Limitations

Key Findings, Threats to Validity, and Future Work



RandomForest Superiority

All classifiers saturated on Mushroom, but **RandomForest clearly separated itself** on the harder Ionosphere task.

Ionosphere (harder, continuous, noisy)

92. 88%

Accuracy — RandomForest

ROC-AUC = 0.98

Mushroom (nominal, highly separable)

100%

Accuracy — all classifiers

Ceiling — no differentiation

Take-away RandomForest is the clear winner on the challenging radar-signal dataset, while easier nominal problems mask model differences.

The Complexity Boundary

More complex models help only where the data demands them.

✓ Complexity pays off

On Ionosphere —

- RandomForest performs best (92.88%, AUC 0.98)
- Feature selection: no stable gain
- More trees: no stable gain

→ **the model helps, but extra tuning did not reliably improve results.**

✗ No added benefit

On Mushroom —

- Classes are already well separated
- Nominal / categorical features
- Even simple models reach 100%

→ **extra complexity is wasted compute.**

Threats to Validity & Limitations

Two issues that affect how far the results can be generalised.

01

Sample Size Constraints

- Ionosphere contains only 351 instances and 34 numerical features.
- Small N creates an inherent risk of overfitting.
- Limits generalisability of models to other radar environments.

02

Class Imbalance

- Ionosphere exhibits moderate imbalance: 64.1% good vs 35.9% bad.
- Raw accuracy alone is insufficient for reliable comparison.
- Multi-metric evaluation (Precision, Recall, F1, Confusion Matrix) is mandatory.

Recommendations & Future Work

Three concrete directions to strengthen the conclusions.

01

Repeated Stratified Cross-Validation

Replace single train/test splits with multiple stratified folds repeated across random seeds.

→ stabilises accuracy estimates and reduces split-dependent luck.

02

External Validation Datasets

Test the learned pipelines on fresh radar / nominal datasets outside UCI.

→ confirms that the RandomForest advantage transfers beyond the two studied benchmarks.

03

Association Rule Sensitivity Checks

Vary support / confidence thresholds and report how the rule set changes.

→ shows that the mined associations are not artefacts of one threshold choice.

Team Contributions

Member ID	Member Name	Main Contributions
3162225	WU Yanping	Dataset Mushroom collection, cleaning, quality analysis and preprocessing.
3162809	ZENG Xinying	Ionosphere data analysis, visualisation and preprocessing.
3163047	HU Ziqi	Classification, association rule mining, feature selection and model workflows.
3163475	ZHANG Jiaming	Model evaluation, cross-validation, significance testing and leakage checks.
3162768	HU Sirui	Report integration, formatting, references, Git management and cross-dataset comparison.

CDS535 Data Mining

Thank you

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